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Project Title:	PASTA: Particle Stabilized Emulsions and Foams		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Comments:	NOTE: Previous affiliation was Lafayette College, Easton, PA until early 2018 when PI moved to Virginia Commonwealth University		
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	NOTE: Continuation grant awarded, 80NSSC21K2080, extending project to 8/29/2023 (Ed., 1/31/22)		
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Task Description:

ED. NOTE January 2022: New grant awarded September 2021, 80NSSC21K2080, extending project to 8/29/2023. Continuation of "Particle Stabilized Emulsions and Foams-80NSSC18K0453" in cooperation with European team, Emulsion Dynamics and Drop Interfaces.
ED. NOTE July 2018: Continuation of "Particle Stabilized Emulsions and Foams," grant NNX15AE98G, due to Principal Investigator James Ferri's move to Virginia Commonwealth University (VCU) in spring 2018.

The overall aim of the PASTA (Particle Stabilized Emulsions and Foams) project is the development of a mechanistic description of the stabilization or destabilization of foams and emulsions, which can be accomplished by various combinations of surfactants, polymers, and particles. This project is continuing as EDDI (Emulsion Dynamics and Drop Interfaces).

Emulsions are liquid-liquid dispersions of significant technological importance. Emulsions consist of droplets of one phase dispersed in a matrix of a phase containing a different liquid. Emulsions are thermodynamically unstable, so that at equilibrium, these liquid phases separate. The existence of kinetically stable emulsions relies in the presence of emulsifiers. These are blends of surface-active species, such as short-chain surfactants, polymers, proteins, and even nanoparticles, which accumulate at the liquid-liquid interface and decrease the energy needed for the creation of new interfaces. These species modify the physicochemical and mechanical features of the interface and counteract destabilization. In contrast, de-emulsifiers can be used to counteract the stabilizing effects present in some emulsion systems (e.g., natural emulsions) to destabilize and separate the liquid phases. These processes are referred to as emulsion breaking. However, a general mechanism of the role of surface-active species in the stabilization of an emulsion has yet to be elucidated. Therefore the development of emulsifier or de-emulsifier formulations are still largely empirical.

The EDDI project, the dynamic behavior and evolution of emulsions will be investigated, including efforts to characterize a) single droplets and b) multi-droplet systems.

a) Single droplets:

•Characterization of the chemico-physical and mechanical features of liquid-liquid interfaces, such as the adsorption dynamics and the dilational rheology, measuring the concerned quantities (dynamic surface tension, dilational viscoelasticity, etc.) on a large dynamic range.

•Develop models for the above features, accounting for the dynamic and structural aspects of adsorption layers. Depending on the system, surfactant exchange between the liquid phases may occur, which also need to be investigated and incorporated in the models.

b) Droplet-droplet & droplet-bubble interactions:

•Investigate and define the conditions for coalescence (drop-drop) and aggregation/incorporation (drop-drop, drop-bubbles) between single droplets, measuring the corresponding rates.

•Investigate the Ostwald Ripening for doublets of droplets, measuring the corresponding rates.

•Correlate the results of these investigations with single drop dynamics.

In addition to the experimental program described above, theory will be developed which describes the behavior above in terms of the molecular and intrinsic constants associated with the materials of which they are composed.

The composition of the consortium of participating teams (USA, Germany, Italy, Greece) is such that all aspects of the formulated problem are covered. The teams demonstrate a complementary expertise in form of the scientific fields and available experimental tools, as well as capacity for developing and applying the relevant theoretical frameworks.

Key experiments in microgravity conditions support the specific aims of this project. The basic processes involved in emulsions and foams, such as destabilization, mainly droplet aggregation and coalescence (destabilization of the film between droplets/bubbles), will be studied in the absence of segregation for model emulsions. In fact, during emulsion and foam destabilization, droplets larger than ten microns are rapidly segregating, making impossible on Earth the investigation of the above processes in controlled conditions. Microgravity experiments will then provide information about the features and characteristic times of these processes to be utilized to parameterize and validate the respective models. Since its inception, the PASTA project has sought to leverage the existing facilities on International Space Station (ISS) Fluid Science Laboratory (FSL). During the past three years, ESA (the European Space Agency) has refocused support in the Soft Matter Dynamics thrust to emphasize the Foam Optics And Mechanics-Coarsening (FOAM-C) facility which was originally developed to study coarsening and stability, respectively, of wet foams (multiple, polydisperse interfaces). As a result of shift in ESA priorities, in 2017 the PASTA team piloted the existing experimental module, FOAM-C, for suitability for emulsion dynamics studies.

DWS (diffusing wave spectroscopy) has been implemented as the main diagnostics of the ESA Soft Matter Dynamics (SMD, formerly FoamC) facility, which has been uploaded onboard the ISS in July 2018, to perform a first set of experiments on granular materials and foams. The presence of the line-camera on the SMD facility will allow the so-called multispeckle analysis, i.e., averaging the evolution of different speckles of scattered light at the same time. With respect to the simple time average done with data from single detectors, multispeckle offers the advantage to reach good statistics in short time, besides enabling to compute time resolved correlation functions, two-time correlation functions, and higher order correlation functions, which are useful to investigate transient states and residual dynamics in arrested systems.

In the framework of the present project, these approaches will be used to follow the temporal evolution of droplet sizes and their dynamics, which is associated to the mechanical properties of aging emulsions, with particular emphasis on low surfactant concentrations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Emulsions have large relevance for applications in a wide spectrum of technologies, processes, and products in different fields including oil and gas, specialty materials, pharmaceuticals, cosmetics, and foods. Both stable and unstable emulsions are needs that are shared across a variety of applications. For example, efficient de-emulsifiers are needed to facilitate the process of phase separation (breaking) in the crude oil emulsions extracted from the wells, while high stability oil-in-water emulsions are required in the final stage of production of the same well or to facilitate pumping high-viscosity oils through pipelines. In the food industry, many products based on emulsified fats or oils (condiments, sauces, frozen confections, and others) require long term stability. However, destabilizing emulsions are key to processing raw food materials such as milk and vegetable oils.
Task Progress:	The Particle Stabilized Emulsions and Foams (PASTA) mission was launched from NASA Wallops Flight Facility in Virginia. A live stream of installation of the experimental module in the Fluid Sciences Lab onboard the International Space Station (ISS) on February 21 was hosted by the European Space Agency (ESA) Belgian User Support and Operation Center (BUSOC). The experimental campaign began around March 1 after calibration and preliminary testing and will end on or about July 15. The team at Virginia Commonwealth University (VCU) has consolidated and internationally-recognized expertise in theory and experiment regarding the study of dynamics of fluid-fluid dispersions, also in relation to the development of experimental techniques and applications to emulsions, foams and nanomaterials. VCU also has complimentary expertise to the Institute of Condensed Matter Chemistry and Energy Technologies / Istituto di Chimica della Materia Condensata e di Tecnologie per l'Energia (ICMATE) team in interfacial rheology, and the University of Parma - Italy (UNIPR) team in characterization of fluid systems using scattering methods. The investigation of aggregation kinetics using dynamic light scattering and its relationship to the stability of emulsions and foams has been in particular one of major interest to the team during the last 10 years. Key experiments in microgravity conditions support the specific aims of this project. The basic processes involved in emulsions and foams, such as destabilization, mainly droplet aggregation and coalescence (destabilization of the film between droplets/bubbles), will be studied in the absence of segregation for model emulsions. In fact, during emulsion and foam destabilization, droplets larger than ten microns are rapidly segregating, making impossible on Earth the investigation of the above processes in controlled conditions. Microgravity experiments will then provide information about the features and characteristic times of these processes to be utilized to parameterize and validate the respective models. Experimental Apparatus: Samples aboard the ISS were contained in Sample Cell Units (SCUs) to be loaded onto the Soft Matter Dynamics Experimental Container (SMD EC) module. The emulsification occurred within the SCU via an oscillating piston. Each SCU contains four samples and the SMD module can hold up to five SCUs, including a reference SCU, for a total of 20 samples. For this project, 16 formulations are examined using an overhead camera, transmission diffusing-wave spectroscopy (DWS), backscattering DWS, and line camera correlation (multi-speckle DWS). Each of these diagnostics are described in the sections below with representative data from the mission. Experimental Diagnostics: Overview Camera (OC) The overview camera is used for a visual inspection of the experiment conduct. The Overview Camera Assembly requires very high resolution and very small pixel size on target (5µm), reason for which a microscope has been designed as camera optic. The camera is focusable on the target via a modified motorized linear scan stage of type PI M110.12S with 50nm displacement resolution. In order to enable moving the camera while maintaining sufficient heat rejection, a thermal braid connecting the camera to the optical subsystem cold plate has been included in the design. The scan stage is also driven by the optics motor driver type EZ4AXIS from Allmotion (same as used for all polarizers). For the Overview Diagnostics, a high-resolution camera type Pointgrey P/N FL3-GE-50S5M-C is being used. This is a monochrome camera with a charge-coupled device (CCD) sensor from Sony ICX655 giving 2448x2048 pixels of 3.45µm size (8.445mmx7.065mm). It has a global shutter with a GigE interface. The optical path between the observed target and the overview camera is a microscope with a very narrow depth of field. This microscope is composed of several lenses that are encapsulated in a metallic housing with polycarbonate windows on both entry and exit points. As a light source for the overview diagnostic, light emitting diodes are used around the target. The sources used are LUXEON Rebel high power LEDs generating a blue coloured light. The overhead camera allows a spatially averaged, but local, analysis of the emulsion. In the frequency space, the images may indicate a coarsening and emulsion structure that is hard to observe in the spatial domain. CorrTectors: Diffusing Wave Spectroscopy (DWS) The SMD EC provides one CorrTector for DWS in transmission and one for backscattering. The CorrTectors are avalanche photo diodes (APD) that count individual photons and send the corresponding trains of logic pulses to a digital Correlator. The correlator collects the logic pulses and computes intensity correlation curves. The resulting g2(t) functions can be transformed into the field autocorrelation, g1(t), during analysis on the ground. Further, the mean squared displacement can be extracted, yielding precise information on the motion of the emulsion droplets in microgravity. We anticipate that the DWS analysis will yield powerful insights on the mechanisms of aging in emulsions on the ground as well as in microgravitational environments. Line Cameras: Speckle-Visibility Spectroscopy (SVS) Line Camera (LC) data consist of several line images, up to approximately 10000, and the number of pixels per line can be specified to be between 1024 and 2048. Each line image is taken at a different timestamp depending on the specified line rate. Effectively, the line cameras aboard the SMD module take photos consisting of a vertical row of single pixels. Due to the low number of pixels, the line camera can achieve faster frame rates than conventional cameras. Using the line camera, spatial configurations and their time rate of change may be assessed. Additionally, the intensity data can be correlated in both time and space to yield similar autocorrelation functions achieved via the avalanche photo diodes in traditional DWS. Some or all of these diagnostics are applied to emulsification protocol and measurement schedules as described in the following experimental protocols A0, A1, A2, and B. Experimental Protocols: Protocol A0: Before starting the recording of the data for this Protocol A0, the uniform illumination and good focus of the top of the Sample Cell will be confirmed based on the real-time stream of overhead camera (OC) images. The diagnostics will be put into streaming mode and recorded on the VMU 2. During the recording, one stroke of the piston

inside the sample cell will be executed. After this piston stroke, the recording will continue for 10 minutes before stopping and closing the record. These 10 minutes can be extended on science team request. This single piston stroke will generate drops that are big enough to monitor clearly, using the OC. Protocol A0 should be performed before any other protocols are executed. Protocol A0 was performed on all samples.

Protocol A1: The first step in Protocol A is to emulsify the sample. The baseline approach is to use a foam generation system (FGS) frequency of 18 Hz and emulsify for a total of 8 minutes, subdivided in 4 FGS activations of 2 minutes with a 1 minute wait time in between. Five minutes of science data will be recorded. If there is a presence of observable intermittent dynamics, then the remaining part of protocol A will be performed. Protocol A1 was performed on all samples.

Protocol A2: The first step will be to emulsify the sample again, using the same emulsifying approach from A1. Next, the observation will be started: 30 minutes of science data will be recorded using all diagnostics. It will be determined by the science team if the playback of the raw line camera (LC) data will be needed in order to perform the full analysis of the run, or if the raw LC can be deleted from the video management unit 2 (VMU2). Protocol A2 was only performed on samples manifesting intermittent dynamics.

Protocol B: The estimated experiment duration ranges from 120h to 360h. To increase efficiency, several Sample Cells (SCU 9, 10, and 11) will run in parallel during Protocol B. Protocol B starts with emulsifying a Sample Cell. Once emulsified, a 6h run of that sample cell is started. This 6h run consists of times of science data acquisition (typically 10 minutes) alternating with wait times (typically 30 minutes) until the 6h run is completed. Once the 6h run of the Sample Cell is performed, 10-minute runs of the involved Sample Cells will be acquired and recorded at the following times: - 6h, 8h, 12h, 18h, 24h, 30h, 36h, and 48h after the emulsification of the Sample Cell. - Every 12h from 48h to a pre-defined estimated duration. Protocol B was performed on all samples.

Experimental Samples and Data:

Each technique is used to construct a time series analysis of the formulations for various total lifetimes. Further in-depth analysis will be performed on the ground after all experiments have been performed.

Ground Based Diagnostics:

The VCU team has focused on developing a second DWS setup for ground-based conditions to enable interrogation of different timescales of dynamic processes.

Since the last report, the VCU team has successfully built a second DWS instrument. The instrument has a rectangular sample geometry and operates with an expanded beam incident to one face. The benefits offered by the new system include: providing an analytical formulation to the electric field autocorrelation function, g1(t), and a larger scattering volume. Test measurements have been conducted and the VCU team is now in the process of calibrating the instrument so that it may be used in conjunction with experiments in microgravity, as well as with the goniometric DWS instrument.

Additionally, in the last report we had observed deviations from the expected transport mean free path length, l*, for SiO2 suspensions in the goniometric instrument. Initially, we assumed deviation was due to optical mixing of scattered and un-scattered light. The VCU team was able to correct these issues with a multi-angle fitting procedure. The multi-angle fitting procedure uses three detection angles, spanning the range offered by the goniometric instrument, measured intensity data from calibration particles, and stochastically simulated intensity data to determine l* accurately. The precision offered by this method allows size determination with accuracy rivaling dynamic light scattering (DLS) and has yielded a publication (in review). Further, the instrument is now suitable to be used to conduct ground-based experiments on similar formulations being studied on the International Space Station.

Bibliography Type:	Description: (Last Updated: 12/03/2024)
Articles in Peer-reviewed Journals	McMillin RE, Orsi D, Cristofolini L, Ferri JK. "Particle sizing in non-dilute dispersions using diffusing wave spectroscopy with multiple optical path lengths." Colloids Interface Sci. Commun. 2022 Jul;49:100641. https://doi.org/10.1016/j.colcom.2022.100641 , Jul-2022